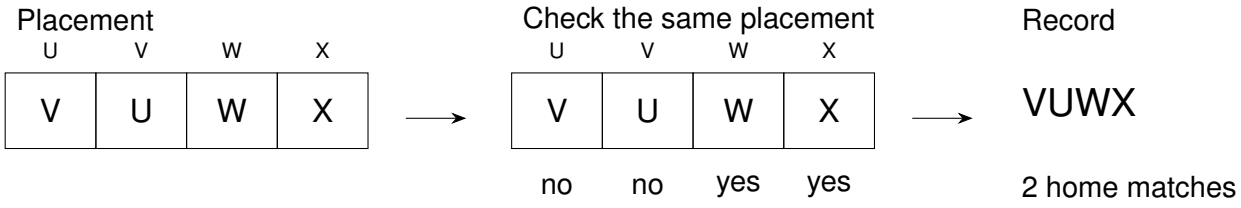
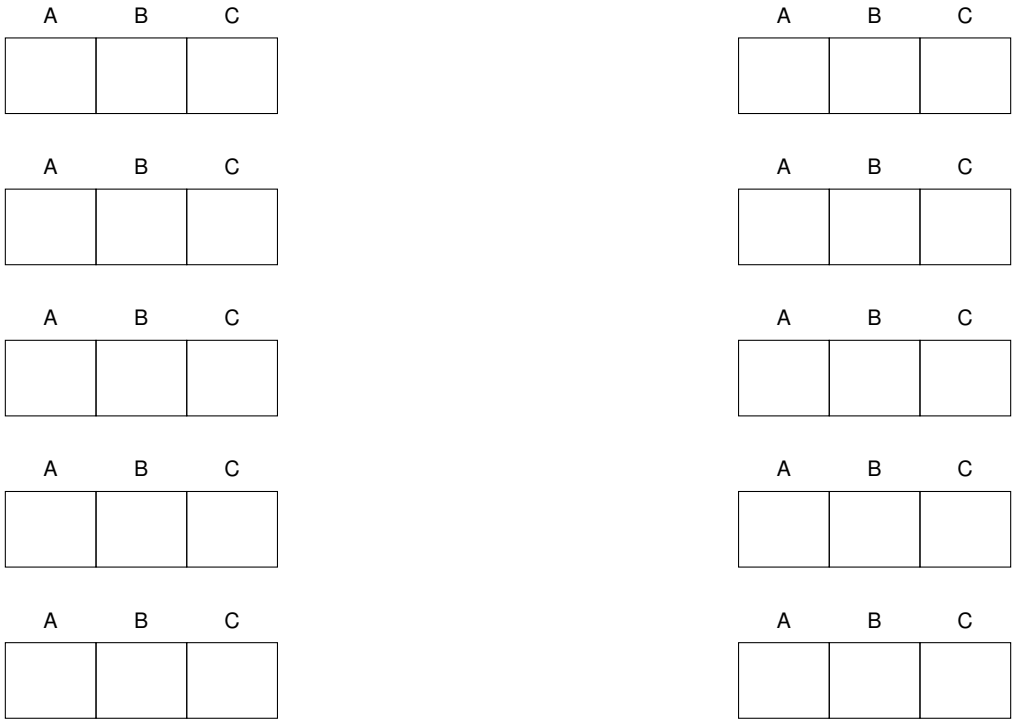
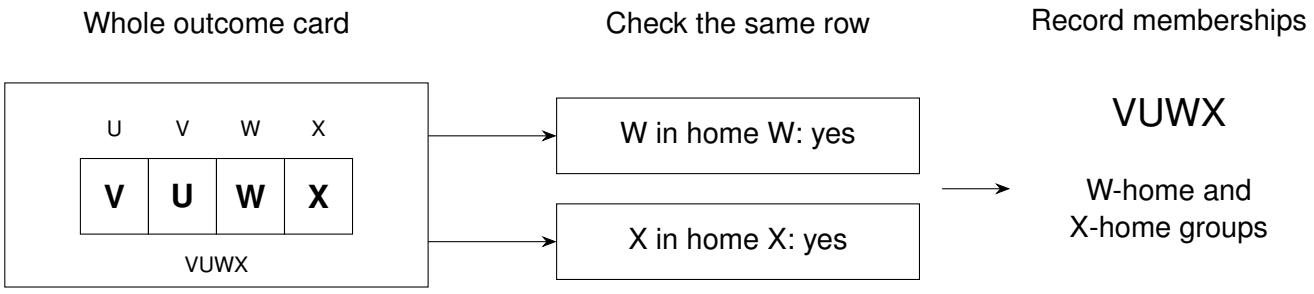
Use a card set and the home strip with the same letters. Use each card once, with one card in every home. Keep the home labels fixed. A *home match* means the card and its home have the same letter. Check a complete row before moving any card. Record the cards from left to right. A row with no home matches is *home-free*.



Problem 1: Find every row of A, B, C with no home matches. Can a row of these three cards have exactly two home matches?



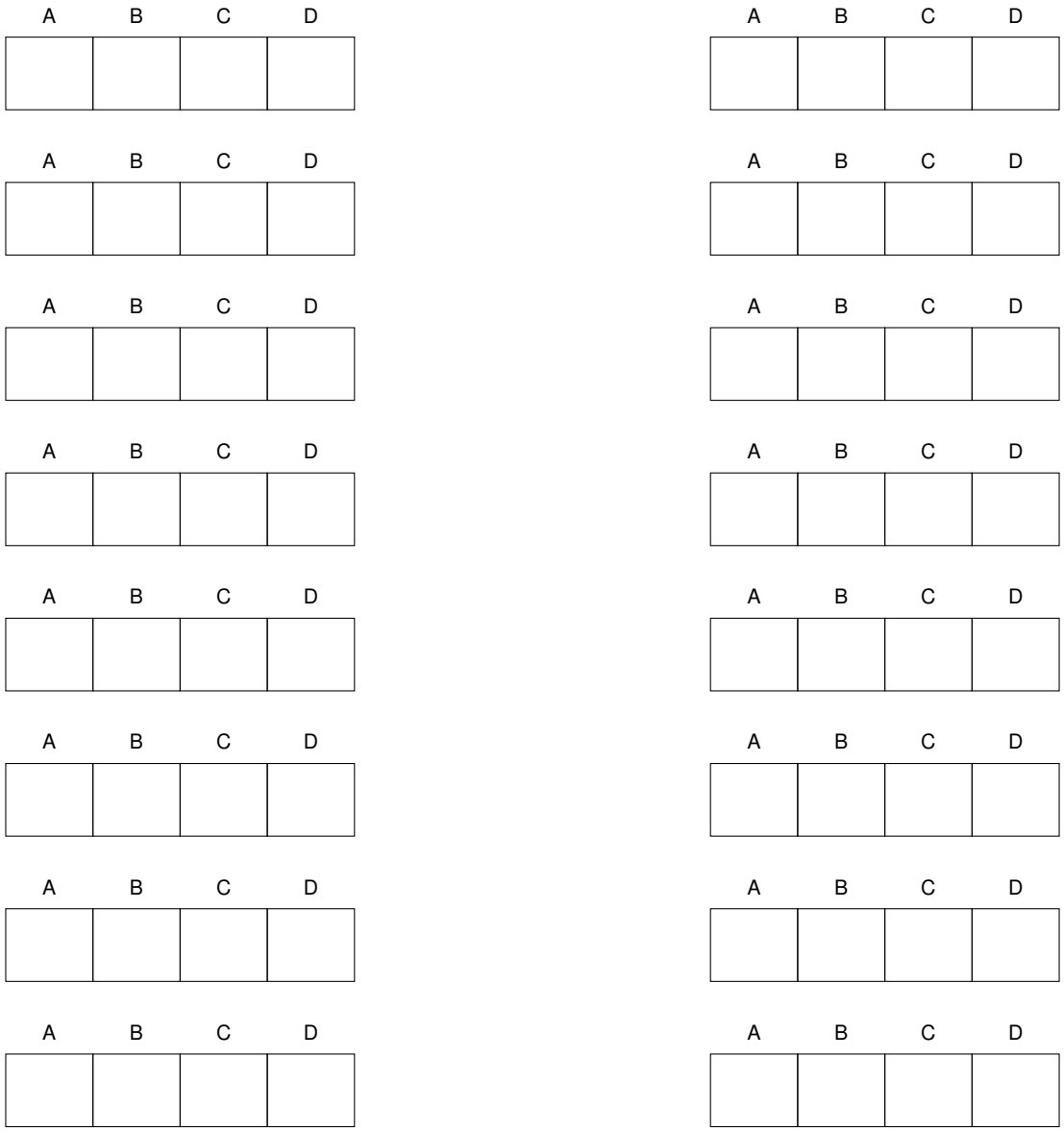
One outcome card is one whole placement.



Problem 2: Use the six A–C outcome cards. Make one group with A in home A and another with B in home B. A card may belong to both groups. How many rows are outside both groups? C may stay home.

A student counts $6 - 2 - 2 = 2$. Repair this count and show what went wrong.

Problem 3: Find every A–D row with no home matches. Make a catalog your table can check for repeated or missing rows.

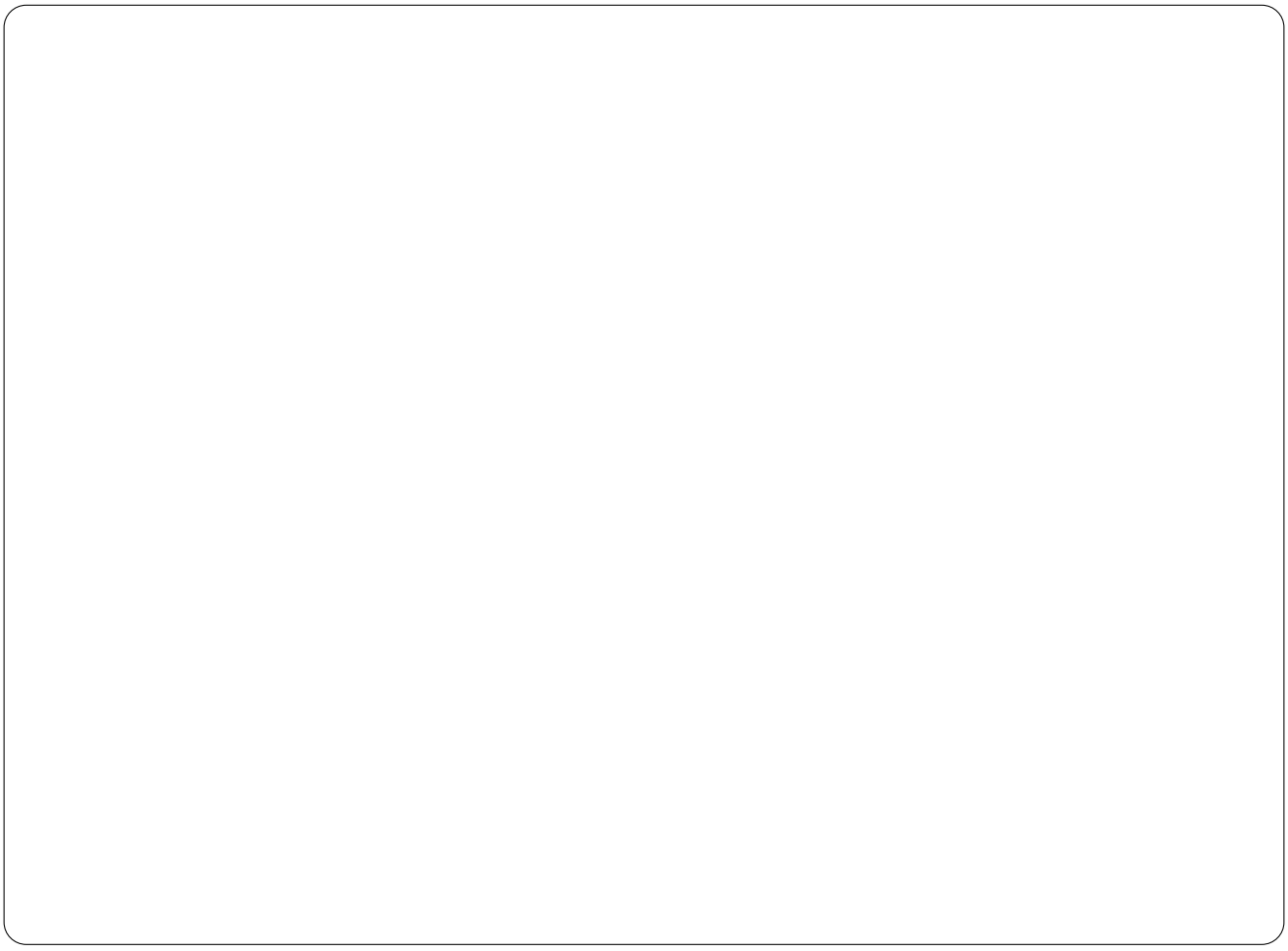


Problem 4: Count the A–D rows with A away from home A and B away from home B. C and D may stay home. Use the 24 outcome cards to check the count $24 - 6 - 6$. How should the count be repaired?

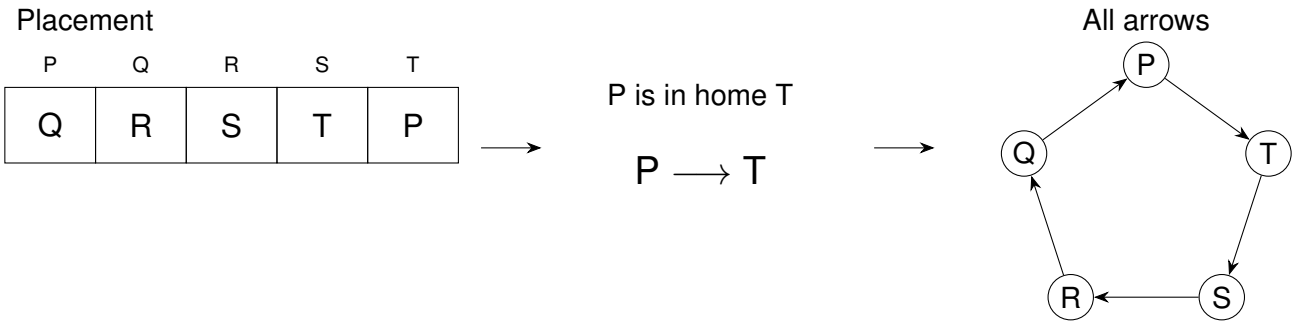
Problem 5: There are four home-groups: A at A, B at B, C at C, and D at D. Each contains six of the 24 A–D rows. A student tries $24 - 6 - 6 - 6 - 6$ to count the rows outside every group. Repair the count using overlaps.

An overlap includes every row with the required matches, even when other cards are also home.

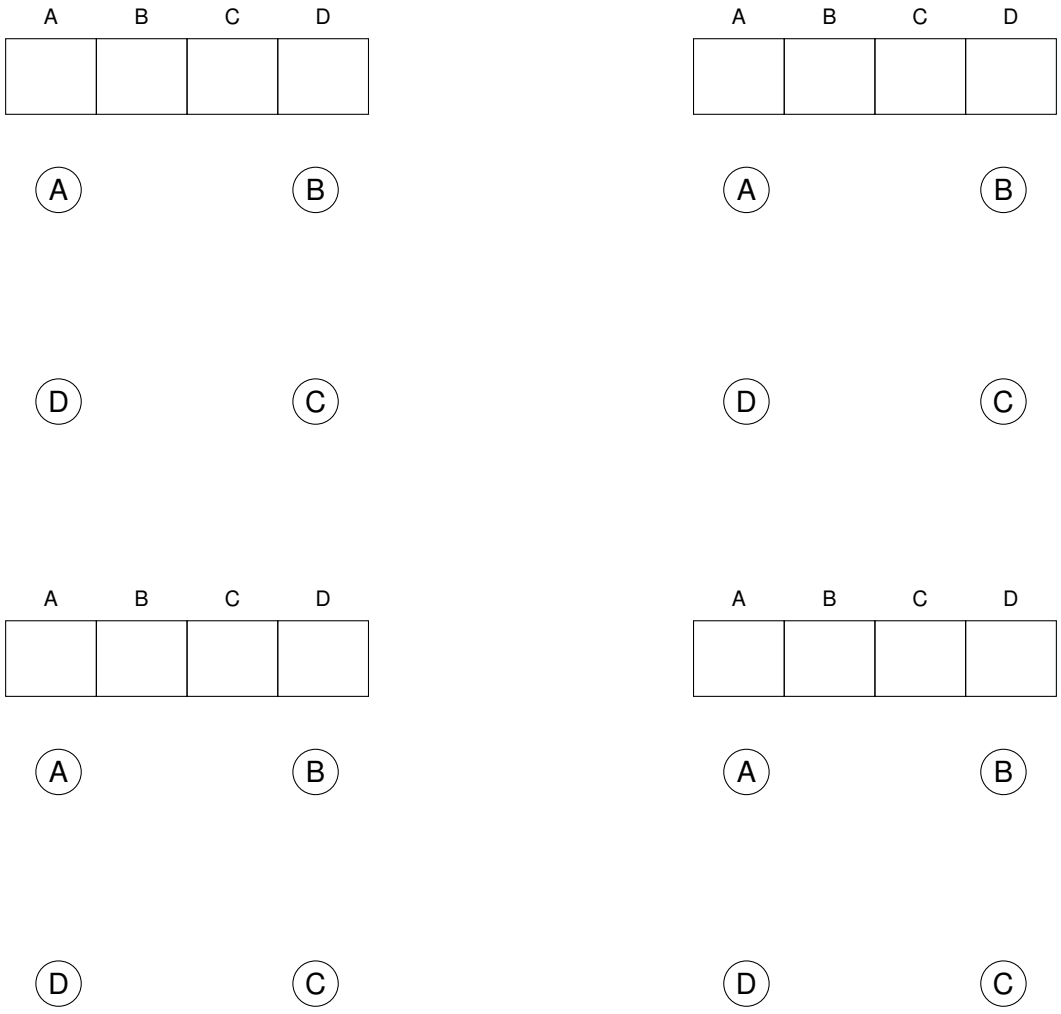
Problem 6: There are 120 A–E rows. Count those with no home matches by correcting overlaps of the five home-groups. Show why your correction keeps each wanted row once and removes every other row.



An arrow points from a card's letter to the home where that card sits.

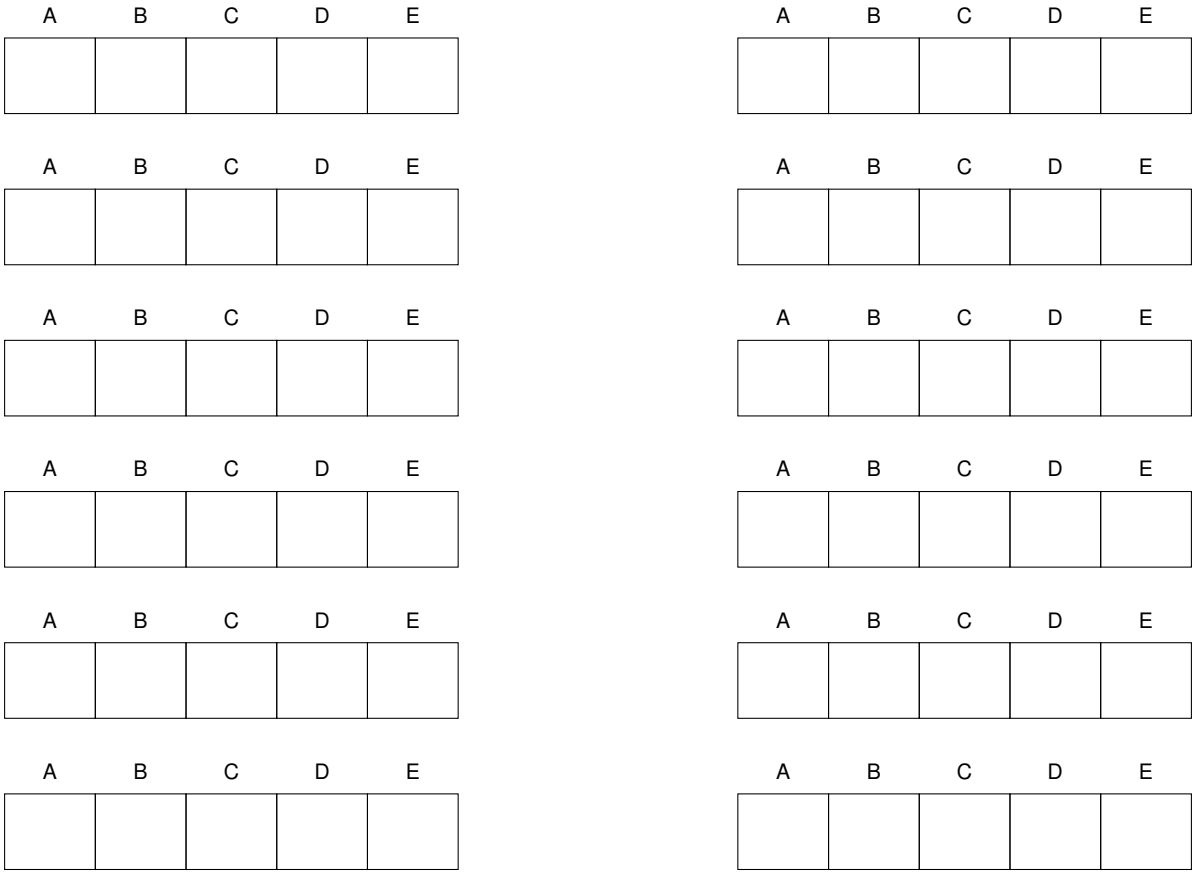


Problem 7: Keep D in home A. Find every A–D row with no home matches, and draw its arrows. Sort the rows into those with A in home D and those with A away from home D. What makes your list complete?



Smaller arrow loops keep their remaining letters.

Problem 8: Keep E in home A. Make every home-free A–E row in two families: A in home E, and A away from home E. How can each family be built from smaller home-free rows? Give constructions you can reverse.



Problem 9: Choose one of homes B, C, D for E. Make a complete catalog of home-free A–E rows with E there. Pool the four catalogs, including Problem 8's, to count all home-free A–E rows. Use the two families to give a counting rule for three or more distinct cards, and explain why each row is counted once.

